# Sleeping Machines — Part II — Methods and machinery

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This part collects how results are produced and checked: evaluation protocols and metric definitions, compute and work accounting, numerical contracts, admission gates, replay drivers, optimizer and precision checks, and the hardware cost model. Sections appear in record order.

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# Appendix B. Native replication scope

Shared S16/P0 order replications: seed7: 44.53%, seed8: 70.31%. The original seed6 screen was75.39%; training-seed variation remains material. Private S16 replication controls and fresh-population confirmation are required for the paired sharing claim; a completed shared-only score cannot pass that gate.

Observed-time paired replications: seed7: 90.23%, seed8: 91.80%. Original seed6 was95.31%, with the exact rank-only paired ceiling50%. These reuse the development distribution and were chosen after seed6; they are training-seed evidence, not independent confirmation.

8 of8 reserved replication pilots are complete in this checkout. Every completed seed is listed in the preceding common-unit tables; pending cells carry no score. The frozen AWS replication/confirmation chain owns the remaining work; no local duplicates.

# Appendix B. Integrated memory repair: accounting prerequisite

Completed short accounting smokes, not quality pilots. Unchanged full native core versus context-addressed outcome slots: H2/d16/depth8, same192 fitting characters/one pass,191 targets, three U64 Adam updates (partial63), c16/lr.002/no warmup, seed6, same core initialization and training race RNG. Frozen cold-state development has128 targets. Both zero-repair initial scores match exactly; every actual fitting operation is traced.

| Model | Smoke dev bpc | Whole fit GFLOPs | Fit MFLOPs/target | Infer MFLOPs/target |
|---|---|---|---|---|
| native | 4.5432 | 0.08838 | 0.4627 | 0.0974 |
| addressed | 4.5384 | 0.08983 | 0.4703 | 0.1016 |

| Model | Core slots | Updates/keys/teacher values per target | Extra slots occupied/capacity | State tensor bytes |
|---|---|---|---|---|
| native | 32 | 16/32/32 | 0/0 | 2,448 |
| addressed | 32 | 16/32/32 | 150/4096 | 21,648 |

Addressed whole-fitting work ratio1.0164×; observed smoke gain0.0047bpc. Three updates are insufficient to assess useful deeper features. Fixed hashed addresses test memory capacity and evidence paths; they do not implement learned context pooling or KV race attention. Native temporal races, evolving messages, receiver key/value separation and unrealized-route surrogate credit remain.

Full-shape addressed/tapped contracts pass exact zero forward/parent gradients, trained causality/chunk invariance, target-weighted normalization and serialized next-Adam predictions, parameters and moments. Context-slot and tap-buffer tensors are now counted and detached. Whole-driver recovery/selection and consistent warm-state comparisons remain admission gaps before long fits; no automatic scale-up.

CPU emulator, 2FLOPs/MAC plus unit-weight specials, all fitting forward/loss/backward/normalization/clipping/Adam; excludes development, RNG, hash/integer operations and traffic. Inference is one accounted target after127 warm tokens, not throughput. Activity columns are derived from fixed code/dimensions. Empty text filler-group NaN in original JSONs is undefined, not a score; preserved beside correction to null in the driver. Cold neural memory differs from fit-prefilled count references. Source: local_deep_feature_preflight_20261002T081500Z.json.

# Appendix B. Value-credit resources and protocol boundaries

| Model | Whole fit GFLOPs est. | Fit MFLOPs/target est. | Cold infer MFLOPs/target | Replay GFLOPs est. | Warm infer MFLOPs/target |
|---|---|---|---|---|---|
| Native full p16/L8 | 1.893 | 0.463 | 0.0974 | 0.100 | 0.0974 |
| Stored projection p16/L8 | 1.924 | 0.470 | 0.1016 | 0.104 | 0.1016 |
| Late projection p16/L8 | 1.933 | 0.472 | 0.0999 | 0.104 | 0.1006 |
| Late p16/L1 | 0.348 | 0.085 | 0.0203 | 0.022 | 0.0210 |
| Late p2/L1 | 0.018 | 0.004 | 0.0012 | 0.001 | 0.0012 |

| Model | Core slots | Extra occupied/capacity | Updates/keys/teacher values per fit target | Context reads/writes per fit target |
|---|---|---|---|---|
| Native full p16/L8 | 32 | 0/0 | 16/32/32 | 0.000/0.000 |
| Stored projection p16/L8 | 32 | 757/4096 | 16/32/32 | 1.000/0.999 |
| Late projection p16/L8 | 32 | 757/4096 | 16/32/32 | 1.000/0.999 |
| Late p16/L1 | 4 | 757/4096 | 2/4/4 | 1.000/0.999 |
| Late p2/L1 | 4 | 757/4096 | 2/4/4 | 1.000/0.999 |

Every column uses the same unit/denominator for all five models. Whole fitting costs are representative first/mature/partial-window estimates from actual forward/loss/backward/normalization/clipping/Adam audits, with admitted losing-value work charged.2FLOPs/MAC plus unit-weight specials; CPU emulator here. Projected clockless costs are separate in the completed JSON ledger. Variable retrieval occupancy and graph reach are not fully enumerated.

Warm evaluation replays all1,024 fit tokens with frozen selected weights, then carries predictive state/time into dev. Hash history and previous-address are cleared at the boundary to forbid an invented cross-split outcome. Dev race noise is coupled to cold scoring. Replay cost is extra work, estimated from first/last replay chunks; it is neither fitting FLOPs nor free access to historical data. Neural slots update causally during dev with frozen parameters.

Late/original full fitting work ratio:1.004×. All direct/core and additional memory work must earn predictive value. No automatic scale-up. The nomination gate requires at least.02bpc cold gains against all four controls and no more than2× original-full fitting work under this common estimate convention.

Hash/integer bookkeeping, RNG, traffic, Python metadata and physical energy remain separate. Cold/replay scores here use2,047 dev targets and cannot be directly juxtaposed with saved8,191-target count/dense references. Original-minimal projection contrast remains open; AWS integrated capacity/exposure work remains independent. Complete work/activity/storage and exact-source checkpoint provenance are retained in local_value_credit_analysis_20261002T090800Z.json.

# Appendix B. Balanced-core fitting and inference work

| Model | Whole fit GFLOPs est. | Fit MFLOPs/target est. | Inference MFLOPs/query | State tensor bytes |
|---|---|---|---|---|
| native_full | 0.5603 | 1.0943 | 0.2424 | 720 |
| tapped_full | 0.6186 | 1.2081 | 0.2750 | 4560 |
| tapped_shallow | 0.1998 | 0.3903 | 0.0978 | 496 |

All rows use 512 query-target presentations as the fitting denominator. Inference includes all 15 episode inputs and query loss. Full prefix forward/backward, losing receiver values, normalization/clipping and Adam are charged. First/last complete optimizer windows supply the whole-fit estimates; routing occupancy is not exhaustively traced.2FLOPs/MAC plus unit specials, including declared delay-index rounding. Actual keys/commits/input events and wall/RSS are saved separately.

Delay taps add bounded per-layer input buffers, trainable delays and projections while retaining native temporal races and private persistent receivers. Zero taps exactly nest native prediction and parent gradients. Seven contracts include actual interrupted-driver and full-shape Adam recovery. The same-width shallow control retains these mechanisms but has fewer layers; initialization also changes, so this is an architecture comparison.

Memory traffic, integer/index bookkeeping, RNG, checkpoint I/O and hardware energy remain outside floating arithmetic. More retention capacity or a valid derivative is not a completed quality advantage. All unsuccessful arms and previous language evidence remain preserved; full source/data/result hashes are in the completed analysis JSON.

# Appendix B. Complete frozen-readout pipeline resource boundary

| Encoder / degree | Total fitting GFLOPs est. | MFLOPs/distinct fit query | Inference MFLOPs/query est. |
|---|---|---|---|
| native full selected / 1 | 0.5918 | 18.495 | 0.2424 |
| native full selected / 2 | 0.6061 | 18.942 | 0.2428 |
| native full initial / 1 | 0.0315 | 0.985 | 0.2424 |
| native full initial / 2 | 0.0456 | 1.426 | 0.2428 |
| tapped full selected / 1 | 0.6543 | 20.447 | 0.2751 |
| tapped full selected / 2 | 0.6686 | 20.894 | 0.2755 |
| tapped shallow selected / 1 | 0.2129 | 6.654 | 0.0979 |
| tapped shallow selected / 2 | 0.2270 | 7.095 | 0.0983 |

Every total includes the entire earlier encoder fit, four-view feature-prefix replay and actual residual-head optimization/feature construction. Initial weights pay zero encoder fitting. The common denominator here is32 DISTINCT fitting queries for every row, rather than differing L-BFGS closure counts. This denominator differs explicitly from the presentation-based table above. Closure target evaluations and exact head work are saved separately. The same feature replay is charged once to each hypothetical standalone arm.

Every inference estimate includes processing the complete prefix plus the chosen query head. Encoder replay/inference estimates use coupled first-quartet operation audits; head construction, loss, backward and optimizer operations are traced directly. Separate retention-probe supervision/work is diagnostic overhead, not free parity training. Raw candidate/traffic/metadata/RNG costs and measured energy remain separate.

Local vector products are supporting primitives, not substitutes for the native temporal core or evidence of learned KV routing. Numerical contracts verify polynomial adjoints, regularized fixed-feature gradients, zero nesting, causal integrated prediction and restored head outputs.

Preserve these negative results beside the subsequent protected-memory hypothesis; no favorable cell is inferred from unfinished training.

# Appendix B. Confirmation fitting and inference resources

| Seed / credit | Whole fit GFLOPs est. | Fit MFLOPs/query est. | Inference MFLOPs/query | Fit wall seconds |
|---|---|---|---|---|
| s7 / joint | 0.2373 | 0.2318 | 0.0534 | 302.3 |
| s7 / local | 0.2314 | 0.2260 | 0.0534 | 315.3 |
| s8 / joint | 0.2380 | 0.2324 | 0.0534 | 307.6 |
| s8 / local | 0.2320 | 0.2266 | 0.0534 | 309.0 |

Every row fits64 distinct queries for16 passes:1,024 target presentations and256 Adam updates, all15 prefix events per query. Whole-fit and per-target columns share that denominator. First/last complete optimizer-window estimates include prefix/native races, all terminal candidate values, joint pair losses where applicable, backward, normalization/clipping and Adam. Measured wall and RSS are separate from FLOPs.

All models have8 native receivers plus27 raw outcome addresses. Each query causes 60 native state commits and14 raw outcome writes. Terminal inference scores2C keys and delivers2 values. Joint training enumerates $C^2$ losses; both training arms inspect C candidate values. C depends on observed occupancy and is recorded in activity traces.

Additional counterfactual arithmetic buys prediction improvement only where completed results support it. A quality advantage at this budget is not an iso-quality compute advantage; shorter or better-controlled local fits have not been optimized. No measured energy, candidate-discovery latency or hardware throughput claim is inferred.

One guarded one-thread CPU job at a time, watchdog active,8GiB available-memory floor. Unique queues, checkpoints, source/data hashes and failed results preserved. This confirmation adds seed evidence for a small structured nonlocal relation, not architectural supremacy or superiority to an unrestricted count-memory algorithm.

# Appendix B. Full and partial-window batched learning admission

| Credit | Initial fit NLL | Selected fit NLL | Whole fit GFLOPs est. | Fit MFLOPs / target est. | Infer MFLOPs / prefix est. |
|---|---|---|---|---|---|
| local | 2.6409 | 2.1145 | 0.107699 | 2.2437 | 0.5917 |
| pairs | 2.6409 | 2.1133 | 0.108205 | 2.2543 | 0.5917 |

| Credit | Workflow seconds | Peak RSS MiB | U16 updates | U8 updates |
|---|---|---|---|---|
| local | 20.821 | 334.8 | 2 | 2 |
| pairs | 20.524 | 334.9 | 2 | 2 |

Each arm uses24 fitting gestures/two fixed passes and8 development targets:48 target presentations/four Adam updates. Both select pass2 and score25% dev accuracy; dev NLL 2.3115 local and2.3070 pairs. These tiny fixed smokes verify learning and resource readiness, not prediction advantage. All optimizer stages have complete operator coverage. Accounting includes candidate values, backward, normalization/clipping and Adam.

Same p16/L2/H2/pool2 integrated architecture:15,523 parameters,8 available receivers, 21 events per prefix,168 scored keys/84 selected state updates/168 candidate values, 720 persistent-state tensor bytes. Independent-clip batching passes forward/state and every-parameter gradient equality plus actual interrupted model/Adam/cursor recovery.

Optional pair credit enumerates the actual final-query two-head outcome losses. Its conditional risk and all derivatives match explicit enumeration; earlier routes retain local surrogate credit. Inference still delivers two hard winners per layer. No exact whole-core gradient or useful-depth claim follows. The bounded matched pilot is next; stronger full-data controls are reported above and are not comparable tiny-fit controls.

One guarded one-thread job at a time; RSS watchdog and8GiB available-memory floor. Theory77; completed admission record local_dvs_batched_smoke_admission_20261002T163600Z.json. All negative full-fit results remain visible; no pending score fills an evidence table.

# Appendix B. Actual-write choice credit: integrated admission

| Credit | Dev accuracy % | Dev NLL | Whole fit GFLOPs est. | Fit MFLOPs / target est. | Infer MFLOPs / prefix est. |
|---|---|---|---|---|---|
| Write-choice + native timing | 25.00 | 2.3113 | 0.143986 | 2.999703 | 0.591695 |

Fixed24 fit/eight dev/two passes,48 presentations/four Adam updates: two U16 and two partial U8 windows. Fitting NLL2.6409 to2.1106; workflow23.041s, peak RSS335.5MiB. This is readiness, not benchmark advantage.

Unchanged native p16/L2/H2/pool2:15,523 parameters/eight available receivers. Fitting pays336 key scores/168 state commits/336 candidate values per target including the complete shadow. Inference retains168 keys/84 commits/168 values over21 events;720 persistent-state tensor bytes.

At one event9 race head/window, substitute the exact conditional expected-loss derivative for both legal delivered values and actual persistent writes. Retain the separate native winner-delay derivative, winning content/state credit and all other local teachers. The obsolete intermediate decoder baseline is unnecessary for categorical enumeration.

Contracts independently enumerate the choice risk, differentiate smooth raw clocks, preserve winning payload and other-head credit, reproduce factual state/logits and actual alternative commits, and verify every parameter-gradient change equals the upstream VJP of the replaced score residual. Interrupted Adam/cursor/RNG recovery and complete operator accounting pass. Downstream timing jumps and other local teachers remain approximate.

Theory84; contracts191100Z and smoke191400Z. All losses summed before gradient normalization, global clipping and Adam; all shadow work charged. No official test or exact whole-model gradient claim. Fixed matched pilot and second-seed gate follow.

# Appendix B. Credit calibration: coordinates and limits

Completed calibration contracts: 5 passed. Common log-rate c and relative logits u define rates exp(c)*softmax(u). The joint winner/time score pulls back to common-clock credit(1-Lambda*T)*(R-b), and categorical credit pi_i*(F_i-R). The choice component is independent of common baseline error; the clock component remains useful and noisy.

Recomposing c=logsumexp(old scores) and u=old scores preserves BOTH the old forward and the old gradient. A coordinate identity does not remove clock noise. Separately learned output maps require an architectural comparison; shared incoming content can still receive both derivatives. Eligible-key normalization and discovery remain paid.

A positive score-space rescaling can oppose parameter descent through coupled Jacobians. The contract gives true parameter gradient[-1,-1] versus rescaled[8,-1], dot product-7. Per-sample unit normalization also biases an explicitly zero-mean scalar estimator. RMS equalization is therefore not a universal calibration rule.

Primary-paper analogues suggest variance-trained legal control variates(RELAX), conditional averaging, phasic updates with constrained route drift(PPG), exposure accounting(MoE) and event

derivative jumps(EventProp). These are distinct repairs. MoE common-logit z-loss would regularize this race clock itself; higher-order optionality needs correct stochastic derivatives(DiCE), not repeated differentiation of detached first-order teachers.

Theory85 mathematical contracts and theory88 linked primary papers; no fitted benefit from these normalization/alternation proposals. Parameter-space covariance and branch exposure on independent-noise fitting probes determine the next single repair. Keep coupled forward messages/races/timing, sparse addressed writes and all existing negative evidence.

# Appendix B. Larger alternative pools: paired-credit readiness

| Credit / pool | Dev accuracy % | Dev NLL | Whole fit GFLOPs est. | Fit MFLOPs / target est. | Infer MFLOPs / prefix est. |
|---|---|---|---|---|---|
| Enumerated2 / readiness | 25.00 | 2.3113 | 0.143986 | 2.999703 | 0.591695 |
| Paired8 / readiness | 12.50 | 2.2989 | 0.414378 | 8.632876 | 0.872348 |

Both24 fit/eight dev/two passes/48 presentations/four Adam updates,U16+partial U8. Paired8 fitting NLL2.5170 to2.1739; 27.533s/349.6MiB peak RSS. These tiny fits establish readiness, not quality rankings, larger-pool headroom or practical superiority.

Paired8 has42,091 parameters/32 available receivers versus15,523/eight for pool2. Per21-event inference prefix:672 scored keys/84 selected commits/672 candidate values; persistent state2,376 bytes. Fitting includes the full shadow: 1344 keys/168 commits/1344 values per target. More state does not mean free key discovery or training.

One full alternative write/suffix forward per fitting window supplies loss-difference credit with recorded proposal propensity. Epsilon.1 gives importance multiplier at most 1/.9; eight auxiliary proposal exponentials/target are recorded separately from arithmetic. Finite2/3/8/64-candidate contracts match independent expected-risk derivatives; pool2 nests enumerated credit exactly. Pool8 legal writes, every-parameter pool2 nesting and actual Adam/cursor/RNG recovery with full/partial windows pass; all operators are covered.

Contracts195800Z, smoke201100Z; theory86/87. Inference and native timing remain unchanged at each shape. No-grad alternatives do not directly teach losing payload maps. Other local teachers and future timing jumps remain approximate. The failed initial195200Z operator-accounting attempt is preserved; covered equivalent subtraction passes a fresh run.

# Appendix B. Fitting-only branch exposure and parameter interference

| Saved seed | Opposed selected choice | Opposed global difference | Mean message cosine | Min sampled/averaged cosine |
|---|---|---|---|---|
| 6 | 0/16 | 16/16 | -0.0481 | 0.999995 |
| 7 | 3/16 | 16/16 | -0.0739 | 0.999970 |

First two fitting prefixes of each saved native local seed6/7 model; four independent whole-history draws, events9/19, both layers, head0. Actual branch writes and full suffixes preserve current first time. Native gradients are compared with an explicit diagnostic blocking ALL race-score paths, including raw-clock score sensitivities. No optimizer or development-label selection; these prefixes do not represent all gestures.

Route/clock-path differences mildly oppose message gradients in both saved models. Averaging both legal message branches leaves the full parameter-gradient direction almost unchanged on these prefixes. Negative inner product alone does not prove harmful interference or justify deleting a chain-rule term. Finite whole-history covariance estimates include changing entering states and clocks; they are not current-node conditional variance.

Independent ordinary-autograd gather contracts reproduce all parameters and full state for six score-blocked factual/legal branches. Detached-probability weighting separates branch derivatives from categorical derivatives. All captured native/branch forwards, backwards and residual VJPs have complete operation coverage; reporting reductions are outside that ledger.

Completed audit203600Z: 273.721s/371.5MiB. Contracts203400Z. No causal explanation of seed7 failure, global exact-gradient claim or benchmark advantage. User-proposed phase offsets have a separate coupled-time hypothesis.

# Appendix B. Counterfactual replay must match the first-time law

| Time-only loss, rates 1/3 | First score gradient | Second score gradient |
|---|---|---|
| True expected loss gradient | -.062500 | -.187500 |
| Incorrect replay choice addition | +.125000 | -.125000 |
| Factual pathwise + wrong replay | +.062500 | -.312500 |
| Correct joint-score expectation | -.062500 | -.187500 |

The first arrival conditioned on either winner has Exp(sum rates) law. Forcing a losing alternative at its individual Exp(its rate) arrival changes both identity and clock. Detached softmax weighting does not turn those combined interventions into exact conditional choice credit. The table uses loss equal to raw arrival time; the first score direction reverses under the incorrect combined estimator.

Four completed contracts include independent analytic expectations and the actual external local-expectation replay wrapper with one time-recording event at four seeds. Forcing the factual winner reproduces the factual delay, while every losing alternative uses its later individual arrival. Existing softmax-algebra and factual-winner tests cannot establish the expected-risk claim. External model, queue and empirical results are preserved; this is a credit-protocol correction, not a fitted quality result.

Correct conditional choice replays retain the factual first time and actual selected memory write. Clock credit needs its own consistent estimator. Factorized winner/first time sampling or the joint likelihood can supply coherent derivations, with all future discrete credit and deterministic branch derivatives included. Do not double-count one sampled time through both likelihood and reparameterized derivatives.

A critic residual correction is unbiased for a declared replay sum when the critic is fixed before its correction subset is drawn. Training on that same subset first can introduce bias: a two-site counterexample gives -1 for a true zero target. Even a sampling-correct critic preserves the base target; it cannot cure wrong replay time laws. Independent conditioning and critic work require separate contracts.

Theory92; contracts212000Z; qualifications beside theory59 sections402/403. Four contracts pass; audited external source versions are recorded. No integrated repair fit, whole-model exactness, guaranteed variance reduction or superiority claim.

# Appendix B. Exact prefix reuse reduces integrated fitting work

| Replay | Dev accuracy % | Dev NLL | Whole fit GFLOPs est. | Fit MFLOPs / target est. | Infer MFLOPs / prefix est. |
|---|---|---|---|---|---|
| full | 25.00 | 2.3113 | 0.143986 | 2.999703 | 0.591695 |
| reuse | 25.00 | 2.3113 | 0.128525 | 2.677607 | 0.591695 |

Detached causal-prefix reuse saves10.738% counted fitting work with identical entire learning curves, predictions, model weights, Adam, cursor and RNG recovery.
Same24-fit/eight-dev/two-pass/48-target native p16/L2/H2/pool2 protocol. Both rows use the same

full-fit and presentation denominators.

Reuse skips repeated alternative-prefix computation while preserving the exact actual-write estimator. This is completed implementation advantage against its full-replay reference, not quality advantage against the strong gesture controls. Snapshot/copy memory traffic and energy remain unmeasured; all retained suffix and optimizer work is charged.

Other-host evidence retained from AWS_PREFIX_REPLAY_FINDINGS_20261002.md and aws_prefix_replay_smokes_20261002T201100Z_analysis.json, with verified parent hashes. Observed13.609 versus12.653 seconds is one concurrent-host observation, not a general latency claim. Exact reuse may support future credit comparisons after matching contracts.

# Appendix D. Evidence and metric definitions

| Metric | Interpretation |
|---|---|
| Bits per character | Held-out negative log probability in base two; lower is better next-character prediction. |
| Accuracy | Fraction of correct class decisions on the declared development or test protocol. |
| Event likelihood | Scores both the next event type and waiting time, including the observed silence. |
| Arithmetic FLOPs | Multiply-add counts as two operations. Tables state whether special functions are separate or assigned unit cost; these conventions must be aligned before forming ratios. |
| Logical operations | The structured-task ledger assigns 2 units per MAC and 1 per other scalar arithmetic/nonlinear operation or estimated sort comparison. |
| Resource boundary | Event-target arithmetic includes required active algorithm work. Memory traffic, queues, indexing and simulator overhead are reported separately where measured. Activity counts alone do not determine total FLOPs. |
| Energy | Measured total joules over an explicit boundary. Operation estimates and CPU timings support work comparisons, but are not joule measurements. |

The evidence is preserved in versioned result summaries with configurations, split identities, learning curves and source hashes. E173/E174 support the language comparison; E61 supports retrieval; E34/E53/E54 support native composition; E41 supports the original periodic computation. E121/E124 establish consolidated arithmetic and its certificate; E123 supplies the new dense controls and E124 the operation ledger. E118/E119/E122/E125/E126 support deep speech, readout and causal-context comparisons; E127–E131 audit credit geometry, hard race boundaries and separate key/value learning; E132 checks a joint race-credit formalism, E133 supplies the language/depth screen, and E134–E135 test whole-value credit and content-selective temporal memory. E136 audits reversible augmented transport and its supervised memory boundary, including twelve-layer query/learning interventions. E137 tests compact memory queries and class-visible credit geometry. E138–E141 examine richer source messages and trainable signed temporal memory, with exact local teacher and initial-nesting contracts. E142 establishes signed-state and first-coalescing identities; E143 tests a larger nonlinear temporal residual learner, and E144 audits simultaneous state/query pooling. E171 reproduces the consolidated screens and selected speech answers, and checks causal input boundaries. E172 records complete training-step arithmetic; E175 checks the generic persistent language stream.

The project theory index contains formal assumptions and proofs. Research findings retain detailed analyses and the full experimental record. The model documentation describes reproducible configurations and operational procedures. This report presents the project, its evidence and its potential.

# 90M protocol, resource boundaries and next decisions

Native pool4 final DEV is1.913993(T128)/1.915118(T256), test1.997194/1.998416. Pool2 DEV is1.963433/1.962040, test2.045381/2.045356. Configuration decisions use DEV; test is reporting-only. Final weights, immutable recovery milestones, source hashes and guarded logs are saved.

Pool4 took10,712.749s, pool2 took7,642.345s, with fitting rates8773.50 and12361.17 targets/s. PeakRSS is recorded in their JSONs. These wall times include their own evaluation and do not establish a paired wall/energy comparison with older dense runs. The saved controls remain better in quality.

Pool4 fitting work is107,606.868GF, or1.195668MF/presentation. Saved LSTM and Transformer whole-fit estimates are36.18x and74.35x greater, but their test scores are1.661 and1.604. These are raw unequal-quality/data work gaps, not achieved quality-matched savings.

Inference work for these trained90M weights remains pending. A prior same-shape winner-only arithmetic trace is a separate diagnostic: trained winner/state/cache/RNG contracts and complete scoring work are required before assigning it to this quality result. Candidate scoring, cache creation and optimizer work cannot be inferred away from selected-write counts.

The completed depth8 fit reaches1.983227 test BPC at127,430.287GF; width64 reaches1.857306 at241,218.897GF. Width64 is the strongest of these four native fits, but remains behind the saved dense controls. Original streaming private full replay and teacher fits continue separately: they preserve chronological state and a different counterfactual estimator. No pending score is predicted. Full causal replay, useful distant-state credit, replication and comparable-quality resources remain open.

Sources: experiments/results/language_batched/aws_language_batched_90M_r2_*.json (completed full fits only), and saved aws_20260929 LSTM/Transformer90M results. The temporal-race, persistent private-state and key/value architectural case is retained.

# Hardware cost model: bytes moved per character

Per-character steady-state inference reads/writes, priced with Horowitz 45 nm anchors (int8, batch 1). Native rows use the exact cached-key winner-only execution (§414); its p96 count agrees with the recorded 1.3 MF/position trace. The 1.888-bpc native model moves 5.8× fewer bytes than the Transformer-256×4 it beats (1.908) and 1.9× fewer than LSTM-512 (1.799, better quality). From pool 2 to pool 32, parameters grow 10.5× and bytes per character 5%: capacity beyond activity at the traffic level (pool-32 quality not measured).

| Model | Params | Weights read/char | State read/char | Bytes/char | µJ/char SRAM | µJ/char DRAM |
|---|---|---|---|---|---|---|
| native p96/d4/H2/U2 | 0.94M | 642K (68%) | 2.3K | 646K | 1.00 | 104 |
| native p64/d4/H2/U2 | 0.42M | 289K (68%) | 1.5K | 291K | 0.45 | 47 |
| native p64/d4/H2/U4 | 0.69M | 289K (42%) | 2.6K | 292K | 0.45 | 47 |
| native p64/d4/H2/U8 | 1.22M | 289K (24%) | 4.6K | 295K | 0.46 | 47 |
| native p64/d4/H2/U32 | 4.43M | 289K (7%) | 16.9K | 307K | 0.48 | 49 |
| LSTM-256 | 0.34M | 338K (100%) | 0.5K | 339K | 0.53 | 54 |
| LSTM-512 | 1.20M | 1199K (100%) | 1.0K | 1201K | 1.86 | 193 |
| Transformer-256x2 (ctx 256) | 1.66M | 1659K (100%) | 262.1K | 1922K | 2.98 | 308 |
| Transformer-256x4 (ctx 256) | 3.24M | 3238K (100%) | 524.3K | 3765K | 5.83 | 603 |

Per-character steady-state inference traffic and priced energy under stated 45 nm conventions; int8 values; batch 1; not a chip measurement. Native rows use winner-only execution with cached key reads (§414, contract tests/test_sparse_inference.py). Source: experiments/results/diagnostics/hardware_cost_model_20261004.json; experiments/hardware_cost_model.py; investment/HARDWARE_THESIS.md. Not a chip, cache or joule measurement.

# Appendix. Tokenized complete work and scaling prerequisites

Synthetic P16/D2/H2/U4/B8/T16, vocabulary50,257; all adaptive tails exercised, first AdamW update. Arithmetic includes factual computation, readout, alternative sampling, replay, backward, clipping and optimizer. Special functions have a separate ledger; random sampling work is unquantified. These are executed steps, not whole-fit extrapolations or hardware energy.

| Scoring | Targets | Step MFLOPs | MFLOPs/target |
|---|---|---|---|
| Full scoring | 128 | 253.930317 | 1.983831 |
| K4 scoring | 128 | 218.255413 | 1.705120 |

K4 arithmetic is14.05%lower in this realization. Replay readout falls47.286432to12.713512MFLOPs. This table is a synthetic first-update diagnostic. Completed actual-corpus8K/P16 and64K/P24whole-fit/inference traces now appear above, with exact trajectory parity and complete arithmetic coverage. The256Kwhole-fit trace is pending; no smaller-fit extrapolation fills it.

Exact actual-driver interruption/resume passed for integrated, sampled-position and uniform-site learners, including all relevant RNGs. Full-score sampled learning equals the original trajectory. Wider decoder tails preserve normalization and initial token priors; default parameterized learning parity and full-width resume passed.

The public reference target reuses the modded-nanoGPT2025-01-26log:3.2774NLL,695,992,320training presentations,10,485,760reserved validation targets. That protocol/context differs from these development rows. EOS/reset/target and layered attention boundaries are source-audited; historical data hashes are absent and no checkpoint is released by this record. No dense reference is retrained.

Prepared AWS8K comparisons remain unrun here, with AWS admission/results unobserved. Completed local paired8Kfits retained credit16 and narrower tails; the64Kcapacity comparison selected P24, followed by the completed256Kfit. Measured-data visualization and source-bound whole-fit costs appear above. The next language fit tests lower learning rate after repeated second-pass deterioration; fixed-recipe scaling and recipe repair stay separate.

Integrated R1 execution: batching vocabulary gathers inside the existing credit windows removes repeated dense embedding-gradient accumulation. All temporal events, hard races, separate keys/values, persistent numerical state, state-gradient boundaries, counterfactual suffix teaching and optimizer updates are retained.

| Lookup | Selected NLL | Final NLL | Train seconds | Engine seconds | Process seconds | RSS MiB |
|---|---|---|---|---|---|---|
| ordinary | 8.045605 | 8.274423 | 149.695 | 152.406 | 153.888 | 565.5 |
| batched | 8.132647 | 8.279827 | 131.433 | 134.053 | 135.533 | 565.2 |

Selected quality gain -0.087041 nats; ordinary/batched training wall 1.139x. Registered development gate (quality retained and at least5% training-wall saving): failed.

One seed; same CPU host/data/updates/selection opportunities and integrated mechanisms. Initial-inclusive quality. Complete measured training wall includes core, alternative replay, head, backward, clipping, diagnostics and optimizer; evaluation/serialization separately in engine wall. Process wall includes acquisition/hash preparation. No fitting-FLOP, energy, Transformer or asymptotic-exponent claim. Exact real-arithmetic lookup equivalence; changed floating-point accumulation can change long hard-route trajectories.

Numerical integration gate: three AdamW updates with carried state, repeated token IDs, EOS resets and two credit windows; maximum forward/state error9.54e-7, gradient error1.20e-7, parameter error1.06e-7, optimizer error7.46e-9; RNG states equal. Evidence aws_integrated_lookup_parity_20261012_v1 and aws_integrated_p24_64k_lookup_decision_20261012_v1.

# Appendix B. Producer-held selection: both rules agree

| Seed | Selection rule | C at984 | Accuracy % | Dev NLL |
|---|---|---|---|---|
| 6 | conditional cv | 0.1 | 54.17 | 1.222593 |
| 6 | producer held | 0.1 | 54.17 | 1.222593 |
| 7 | conditional cv | 0.1 | 53.65 | 1.177975 |
| 7 | producer held | 0.1 | 53.65 | 1.177975 |

| Seed | Original core fit GF | Feature replay GF | Port-check replay GF | Solver/grid GF |
|---|---|---|---|---|
| 6 | 2.285696 | 0.888889 | 0.299717 | Unknown |
| 7 | 2.285696 | 0.888889 | 0.299722 | Unknown |

Use the two saved256-fit native pilot encoders after four fixed passes, without development checkpoint selection. Conditional three-fold decoder CV uses producer-seen fitting labels. Producer-held selection fits on those256 examples and scores728 fitting examples whose labels the producer never saw. Both rules choose the strongest nominal C.1 in both seeds; no selection disagreement or causal selection failure is demonstrated here.

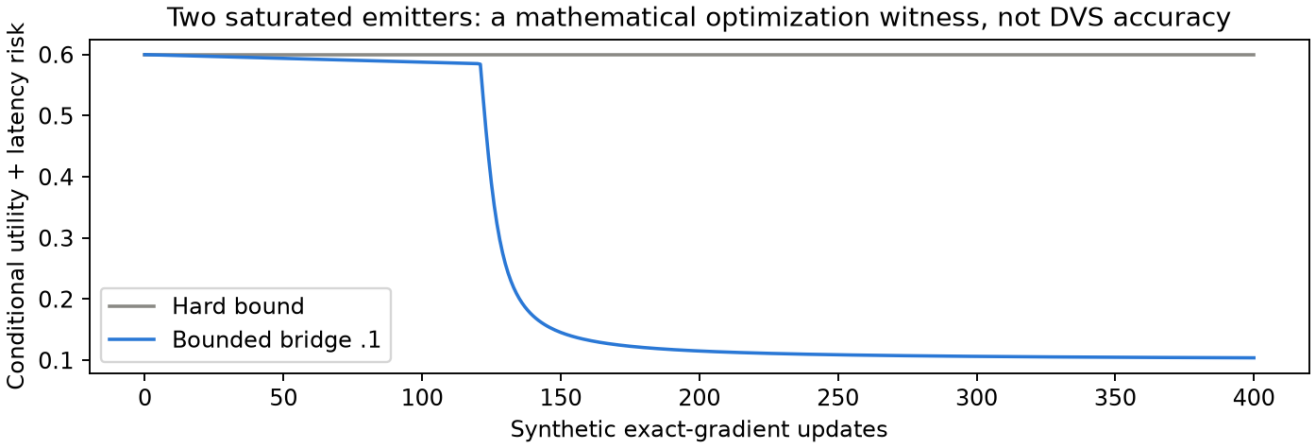
Three mean-L2 coefficients are invariant across fold sizes: lambda=1/(984C), solver C=1/(n lambda). The selected affine heads are then refitted on all984 fitting examples. Final54.17%/1.222593 and53.65%/1.177975 remain weak. Because the new heads see984 labels versus the original pilots256, this is neither an equal-data benchmark nor evidence of practical advantage.

Four contracts include a constructed feature-selection confidence counterexample, sample-size invariant regularization, reproduction of fixed-pass native probabilities and folding scaled affine coefficients into the original native head. Every non-head parameter is bitwise preserved; batched and serial native probabilities match. Parameter count and the inference architecture remain unchanged.

Theory96. The mathematical counterexample establishes a possible failure, not its occurrence in these pilots. Original encoder fits, feature replay and head-port verification replay are separate costs; solver/transformation/traffic/ energy work remains unknown. Same reused192 development examples; no official test or new main model. No guessed global regularization fit is admitted from this result.

# Appendix B. Score-bound exposure and an exact gradient trap

| Seed | Encoder | Evidence | Capped races % | Expected capped winner % | Query races % |
|---|---|---|---|---|---|
| 6 | initial | Cap proxy | 0.00 | 0.00 | 0.00 |
| 6 | fixed_pass4 | Cap proxy | 9.53 | 9.53 | 32.62 |
| 7 | initial | Cap proxy | 0.00 | 0.00 | 0.00 |
| 7 | fixed_pass4 | Cap proxy | 7.43 | 7.43 | 25.78 |
| 6 | initial | Raw-confirmed | 0.00 | 0.00 | 0.00 |
| 6 | fixed_pass4 | Raw-confirmed | 8.56 | 8.56 | 34.38 |
| 7 | initial | Raw-confirmed | 0.00 | 0.00 | 0.00 |
| 7 | fixed_pass4 | Raw-confirmed | 6.99 | 6.99 | 25.00 |




*local score bound evidence 20261002T235800Z trap*

The43,008-race eight-history profile gives cap-occupancy proxies; the5,376 one-history score-pair decomposition independently confirms raw saturation using a2e-5 margin. Cohorts/noise scopes stay separate. No sampled race has both candidates saturated: the two-bound trap below is a mathematical witness, not an observed native condition. Late query cap exposure is stronger than all sites.

Theory105 exact conditional utility+bounded computational-latency risk: two raw emitters13/13,400 updates/step1. Hard remains .600003; bridge .600004 to.103999. A finite hard-map hop improves risk, proving an optimization flat cell rather than a representational impossibility. No stochastic estimator noise/native fitting/quality gate;800 scalar updates paid. Saved-array study .332s/245,116KiB; original fits retained, total audit FLOPs unknown.

# Appendix B. Query-only native admission: exact arithmetic saving

| Seed | Dev acc % | Dev NLL | Whole fit GF | Fit MF/presentation | Old infer MF | Query infer MF |
|---|---|---|---|---|---|---|
| 6 | 66.67 | 0.890983 | 4.218015 | 0.535825 | 0.138119 | 0.135259 |
| 7 | 61.98 | 1.056934 | 4.218015 | 0.535825 | 0.138007 | 0.135147 |
| 8 | 69.27 | 0.920744 | 4.218015 | 0.535825 | 0.138035 | 0.135175 |

Three saved coarse4/.25-clock native producers,984FIT/eightpasses/7872 fitting presentations and192DEV. All576 paired prefix logits and ALLstate checks are bitwise identical. Only four unused intermediate affine classifiers are removed; clocks, races, keys, messages, persistent writes and terminal query remain. Original weights/fitting work and predictions are retained.

2860 counted operations per prefix,2.07%, are saved under full ATen coverage. Three counterbalanced wall repeats show the Python wrapper about2% SLOWER. Arithmetic saving is supported; practical speed/supremacy is not. This explicit-query inference-only helper cannot replace ongoing-label or silence-supervised streams without a separate contract.

65.025s/465,880KiB; target/repeat/query/parameter/training-rejection checks pass.8available units,4writes/event,all core scoring still paid. Same fit denominators/units and first11-prefix inference convention for both implementations.2FLOPs/MAC plus unit specials; raw preprocessing, traffic, RNG and measured energy remain separate. Verbatim former manual section preserved in appendices/aws_query_only_history_20261002T233000Z.md.

# Appendix B. Bounded score bridge: integrated training contracts

Eight double-precision contracts pass before fitting: alpha0 logits and all parameter/content gradients exactly nest the old batched path; positive bridge matches independent sequential factual and full first-time alternative replays; real addressed writes differ; full-route gradients, normalization, clipping and Adam agree. Partial-window model/Adam/RNG/cursor continuation is bitwise exact. Full/partial fitting stages and selected-value native inference have complete operation coverage. Explicit zero gradients in batch and absent unused sequential gradients are equivalent; any nonzero mismatch fails.

| Variant | Dev acc % | Dev NLL | Whole fit GF | Fit MF/presentation | Infer MF/prefix | Presentations |
|---|---|---|---|---|---|---|
| Hard | 31.25 | 2.281937 | 9.769170 | 152.643283 | 0.333083 | 64 |
| Bridge .1 | 31.25 | 2.281964 | 9.827467 | 153.554179 | 0.335771 | 64 |
| Saved p16/L2/all | 58.33 | 1.141794 | 1040.396781 | 132.164225 | 0.591779 | 7872 |
| Saved p16/L4/k8 | 55.73 | 1.330390 | 214.394078 | 27.235020 | 1.068297 | 7872 |

Matched smoke arms: seed7,p8/L4/H2/pool2,32FIT/16DEV,two passes/64 presentations,U8/lr.003,eight Adam updates,all168 races and336 alternative shadow lanes per21-event prefix.7899 parameters,16available receivers, 16key scores/eight actual writes per event,720live state bytes in the 11audited DEV prefixes. Keys/values, temporal computation, source carry and sparse persistent writes remain; fitting computes all candidate values.

Both arms learn on FIT: initial/final NLL2.829119/2.037279 hard and 2.829112/2.039687 bridge. Equal31.25% small DEV accuracy; bridge NLL is worse by.00002646. Bridge adds.5967% counted fitting and.8070% inference work. Crucially, neither initial nor selected smoke model saturates a single one of5376 FIT races (max raw score2.59 initially/2.30 selected). This is integration evidence; it does not test saturation repair or nominate a larger fit.

Contracts9.065s/344424KiB; first missing-versus-zero comparison failure retained. Smokes61.158/64.604s,373200/374120KiB under one-host guard. Saved references use984FIT/192DEV/eightpasses/7872presentations: different width/data/quality; raw whole-fit gaps106.498x/21.947x versus hard smoke are unsupported as efficiency advantages. Same units and target denominators within each column;2FLOPs/MAC plus unit specials, first11 DEV inference convention. Preprocessing/traffic/RNG/energy separate; no official test.

# Appendix B. Deep causal language: admission only; ten-million fits pending

| Family | Params | Fit targets | Initial BPC | Final BPC | Whole fit GF | Fit MF/char | Infer MF/char |
|---|---|---|---|---|---|---|---|
| private | 54907 | 1024 | 5.311638 | 4.832740 | 0.463220 | 0.452363 | 0.097376 |
| depth | 22687 | 1024 | 5.273659 | 4.807838 | 0.461066 | 0.450260 | 0.097376 |

Both integrated native depth8 families pass causal token/teacher parity and actual private-state/Adam/cursor/RNG/partial-window recovery contracts. These1025-character/1024target,513character/512target DEV,onepass,credit16/ U256 learning smokes use the ORIGINAL counterfactual teacher, not corrected full replay. Model key/value separation, sparse receiver writes and physical time evolution remain.32available units,32scored keys/16writes perchar; 512target DEV retains2448bytes addressed state in both families. Four optimizer updates, all core fitting work charged.

Private BPC5.311638 to4.832740,shared5.273659 to4.807838 are admission checks, not language advantage or evidence that deeper features generalize. AWS reserved coordinator now owns private/shared first10M-character FIT models,onepass,disjoint1M DEV at90M,seed7,credit16/U256/lr.002/warmup4096. No completed10M quality cell is filled here. Gesture

replay nominations failed confirmation; language teacher runs remain distinct controls.

The corrected language choice return must sum ALL downstream token losses after the forced route, actually changing private memory at the factual FIRST time. A detached chunk omits later-chunk credit. Full replay at T16/L8/H2/pool2 costs512shadow lanes/8192shadow events perchunk; causal suffix snapshots can reduce about2x, not erase the quadratic work. A future language port needs full-parameter sequential/batched sum-return contracts, causal intervention, actual recovery and complete paid work.

Completed smokes250.174/251.253s,peak<534MiB; bounded long jobs stay in AWS tmux/coordinator with reserved2CPU slots and8GiB floor. Table uses CPU emulator arithmetic plus unit-special convention for both models; projected physical-event numbers remain separate. Credit graph freed each16target microchunk, persistent state crosses updates. DEV passes/ RNG/traffic/energy outside fitting FLOPs, not zero. Source snapshots frozen; all pending runs retained as pending and independently owned.

# Appendix B. Corrected causal language replay: sixteen contracts pass

| L8/p4 family | Params | Entering events | Entering bytes | Final bytes | Parameter grads | Shadow lanes |
|---|---|---|---|---|---|---|
| private | 4411 | 2 | 1040 | 1320 | 384 | 96 |
| depth | 2071 | 2 | 1000 | 1320 | 174 | 96 |

A new sibling language helper emits EVERY token prediction from nonempty detached native receiver memories/arrivals/source context. For each race at event t, replay each alternative at the factual FIRST time with common future randomness, real private memory writes and coupled suffix clocks/messages; stopped-gradient utility sums losses t..T-1. Categorical local-expectation credit is added to factorized common-clock and realized-content derivatives. Original hard score map retained, no bounded bridge/calibration, token target in input or per-position KV bank.

Both private and depth-shared L8/H2/p4/pool2 double models pass EVERY-parameter sum-return gradient comparison with independent sequential full-write replays, all private state/arrivals/context comparisons, original-teacher/factorized exact forward-state parity and variable2/3 token lanes. Losing routes preserve factual first time, change actual suffix predictions and preserve preceding predictions. Future observed tokens and changed target labels cannot affect earlier factual output.

Actual partly accumulated gradients, native state, Adam, cursor and RNG recover bitwise exactly; independent sequential summed gradients agree after target normalization/clipping/Adam. All factual/shadow/backward and optimizer operations covered.96shadow lanes/288shadow events per three-target case are charged. The port closes a numerical mechanism gap; no benchmark fit or deep-feature advantage is inferred.

Theory108;23.973s/361084KiB,16contracts. State bytes are actual double-precision numerical states, not the production float ledger. Truncating/detaching entering credit still omits later-chunk derivatives and gives no whole expected-stream gradient theorem. Existing AWS10M original-teacher controls and their immutable sources stay distinct; no new10M quality cell or claimed language replay gain.

# Appendix B. Production-size language replay: paid resource admission

| p16/L8 case | Targets | Whole step/fit GF | Fit MF/target | Infer MF/target | Shadow lanes |
|---|---|---|---|---|---|
| private/full16 | 16 | 1.084409 | 67.775587 | 0.097256 | 512 |
| private/partial3 | 3 | 0.040100 | 13.366694 | 0.097256 | 96 |
| depth/full16 | 16 | 1.083893 | 67.743327 | 0.097247 | 512 |
| depth/partial3 | 3 | 0.039584 | 13.194644 | 0.097247 | 96 |
| private/teacher smoke | 1024 | 0.463220 | 0.452363 | 0.097376 | 0 |
| depth/teacher smoke | 1024 | 0.461066 | 0.450260 | 0.097376 | 0 |

Private54907/depth-shared22687 parameters,H2/pool2,32available receivers,32key scores/16actual writes per observed token. A full16target chunk executes512shadow lanes/8192shadow events; partial3 executes96/288. Actual original temporal state and hard races remain. Both full factual primal outputs/state match independent sequential native computation. Four prespecified suffix returns per family match sequential alternatives at factual first time; all eight earlier-loss/end-RNG checks pass.

After each real full-step Adam update, save model/nonempty optimizer/ private state/cursor/RNG and recover the next actual three-target partial update bitwise exactly. Total four test updates plus two recovery repeats are paid. First-state2376/2304bytes, final2448bytes, float32. Every factual/shadow/backward/normalization/clipping/Adam/native-inference operation has complete coverage. Numerical admission succeeds86.859s/ 429912KiB under guarded one-thread tmux,~11GiB host memory available.

Full replay costs67.78/67.74MF per target here versus saved original teacher smoke about.45MF. Tables show whole executed-step or whole fit work and the corresponding actual target denominator in the SAME units. Synthetic16/3target correctness strings and saved1024target text8 smokes have unequal data/quality/update context; raw work gaps are not efficiency claims. Native sparse inference stays~.097MF/target but does not remove counterfactual learning work. No DEV/test/BPC or fitted replay gain.

Theory109; production tests cover full execution and selected returns/recovery. Full EVERY-parameter sequential-return comparison is the prior T3/p4 double contract, not relabelled full T16 equality. TwoFLOPs/MAC plus unit specials; shadow-state export and taps are paid diagnostic execution/traffic. Development passes, physical traffic/RNG/ energy remain separate. Whole diagnostic-audit FLOPs unknown, not zero. No unchanged language training promotion follows from resource admission.

# Appendix B. Stateful language replay accumulator: chronological RNG retained

| L8/p4 family | Params | Targets | Chunks | Shadow lanes | Shadow events | Partial LR | Live bytes |
|---|---|---|---|---|---|---|---|
| private | 4411 | 3 | 2 | 96 | 160 | 0.0015 | 1320 |
| depth | 2071 | 3 | 2 | 96 | 160 | 0.0015 | 1360 |

New sibling RNG-state kernel/helper and ReplayAccumulator preserve the original native stream randomness cadence: enter a chunk from its actual Torch RNG state, replay all alternatives from that SAMEstate, and advance the real stream only to the factual end RNG. No extra seed draw, repeated chunk seed or input/label/index randomness channel. Private addressed state crosses chunks/optimizer updates while its credit graph detaches each microchunk. Physical races/message evolution and original hard score map remain; numerical shadows are fully charged.

Ten double L8/H2/p4/pool2 private/shared contracts pass. Every factual prediction and ALLstate match original chronological-teacher forward before a delayed optimizer update; end RNG is exact. Independent sequential actual-write all-target returns match every accumulated parameter gradient over two-plus-one target microchunks. Save partly filled gradients/private state/Adam/cursor/RNG and all replay counters: recovered predictions/state/gradients and next actual update are bitwise exact.

Target labels change pending gradients but leave both chunks' pre-update factual predictions/state/RNG identical. Target-weighted normalization,clip1,warmup total3/4,partial lr.0015 and Adam match independently summed sequential credit. Complete actual shadow/backward plus normalize/clip/warmup/optimizer operation coverage.96shadow lanes/ 160shadow events across the2+1chunks are paid; this is not the288 events of one unbroken three-target credit chunk. Different boundaries are explicitly different training objectives.

Theory113;14.510s/345160KiB. Correctness-test optimizer steps, no trained data/DEV/test/quality claim. Successfully contracted note108 seeded sources are preserved; new stateful siblings are frozen too. Existing AWS10M original-teacher drivers and checkpoints unchanged. A new fitted driver must use this credit in BOTH traced and untraced windows, save replay counters and preserve accounting/chronological controls; merely swapping an accumulator into an old traced loop would omit replay in traced windows. No fitted comparator launched here.

# Appendix B. Replay fitting driver: interrupted learning recovers exactly

| L8/p4 family | Params | Targets | Updates | Whole fit GF | Fit MF/target | Infer MF/target |
|---|---|---|---|---|---|---|
| private | 4411 | 8 | 2 | 0.008089 | 1.011185 | 0.009850 |
| depth | 2071 | 8 | 2 | 0.008013 | 1.001668 | 0.009810 |

New sibling benchmark driver ALWAYS calls the stateful full-write ReplayAccumulator. Operation tracing wraps that same call, preserving all-target shadow returns in both audited and ordinary windows. Twelve end-to-end private/shared depth8 contracts pass. Synthetic eight-target fits use two-target credit chunks, four-target optimizer windows, target-weighted normalization, clipping and warmup. This closes the traced-loop omission hazard identified in Theory114; the existing AWS original-teacher controls remain unchanged.

Interrupt after the first microchunk: two targets of pending gradients, zero updates, live private memory and an unfinished work trace. Interrupt separately after the first completed update. Both resumptions reproduce EVERY learned weight, Adam field, remaining gradient, state, RNG, counter and cursor bitwise, together with all output curves, DEV selection and fitting work. Tracing and untraced learning also agree bitwise; untraced runs explicitly carry no whole-fitting work estimate.

Changed source hashes, settings or input data refuse recovery; completed outputs refuse overwrite. All factual/shadow forward and backward, normalization, clipping and optimizer operations have complete numerical accounting coverage. The eight-target row pays 256 shadow lanes and512 shadow events across four detached chunks. Inference is the original cold native selected-value prefix. Whole-fit and per-target columns use the SAME actual eight-target denominator for both families; synthetic correctness fits provide no text8 quality comparison or benchmark advantage.

Theory114;83.791s/346016KiB. All numerical arms ran serially inside one guarded job with an8GiB host-memory floor. Original races, score clamp, key/value separation, deep persistent state and chronological noise are retained. No larger credit horizon, new timing operator or Transformer fit is admitted by these contracts; a real-data replay fit still requires its own frozen protocol and measured throughput budget. Older AWS DVS results retain their exact original source hashes: the shared regularization driver later evolved, so publication validates its archived original bytes from Git rather than requiring the current driver. Stored results and frozen report modules are unchanged.

# Appendix B. Factual-winner reuse: measured work and numerical limits

| L8/p16 family/arm | Params | Whole fit GF | Fit MF/target | Infer MF/target | Shadow lanes |
|---|---|---|---|---|---|
| private/original | 54907 | 1.084409 | 67.775587 | 0.096786 | 512 |
| private/reuse | 54907 | 0.546353 | 34.147058 | 0.096786 | 256 |
| depth/original | 22687 | 1.083893 | 67.743327 | 0.096786 | 512 |
| depth/reuse | 22687 | 0.545837 | 34.114799 | 0.096786 | 256 |

Every sampled factual winner already supplies its true downstream loss. Forcing that SAME receiver at the SAME first time, with the same entering state/future draws and actual write, reproduces the factual trajectory. Reuse this detached return and simulate only different outcomes. The categorical local expected-return objective and every earlier score/producer VJP are identical in exact arithmetic; native clock/content derivatives and inference stay unchanged. At two candidates this removes half the shadows; it changes neither credit horizon nor candidate support.

Theory115:28private/shared double depth8 contracts for pools1/2/4 match EVERY parameter gradient against both old batched and independent sequential enumeration. ALLfactual-winner replay outcomes agree, winner-recording primal/state/RNG/pathwise gradients nest bitwise, causal token/label and exact pending-gradient/Adam recovery pass. Production rows above each pay ONE synthetic sixteen-target update from the SAME nonempty state/parameters/RNG:512→256shadow lanes and 8192→4096shadow events. Actual normalization/clip/warmup/Adam and all factual/shadow backward are included, with complete operator coverage.

private: counted fitting work -49.617%; gradient relative L2 9.96e-07, max absolute 1.44e-05; 3 parameter tensors miss the coordinate tolerance; actual Adam-update relative difference 0.000257. Production numerical gate FAILS. depth: counted fitting work -49.641%; gradient relative L2 9.73e-07, max absolute 1.79e-05; 3 parameter tensors miss the coordinate tolerance; actual Adam-update relative difference 0.000594. Production numerical gate FAILS.

Theory116;98.060s/399744KiB. Predeclared float32 coordinate rtol3e-4/atol3e-6, global gradient relative error<=3e-5 and Adam-update relative error<=.005 are retained. Initial strict run stopped at a key-read coordinate mismatch; the completed audit reports all gates without weakening them. Nonempty Adam/private state/pending three-target recovery and next partial update are bitwise exact. Tables use the SAME16target denominator and2FLOPs/MAC+unit-special convention for all arms. Extra recovery steps are separately paid diagnostic work. No data-fit BPC, energy, physical projection or benchmark superiority claim; existing AWS10M teacher/factorized/full-replay sources and queues remain untouched.

# Appendix B. Production precision: equivalent gradients and every shadow route agree

| Family | Double reuse error | Float32 old error | Float32 reuse error | Old/new decisions | Mismatches |
|---|---|---|---|---|---|
| private | 1.44e-15 | 1.49e-06 | 1.6e-06 | 131328/65792 | 0 |
| depth | 2.73e-15 | 2.12e-06 | 2.1e-06 | 131328/65792 | 0 |

Same represented float32 p16/L8/H2/pool2 weights and nonempty private memory as Theory116, promoted to double without reinitializing the model. Same synthetic sixteen-target chunk and entering RNG. Compare EVERY native parameter gradient of original full enumeration and winner reuse at both precisions. In double the complete production gradients agree to1.44e-15/2.73e-15 relative error; factual logits/state are bitwise identical. All caller and factual-end RNG states match.

The table compares each float32 implementation against its OWN double program. Both show roughly one-to-two parts per million global gradient error. ALL131328 original and65792 optimized factual/shadow race decisions per family agree across precisions; there is no branch crossing to

explain away the comparison. The prior tight-coordinate failure therefore coexists with exact mathematical estimator equivalence and floating arithmetic error in BOTH implementations. Prior failed gates remain recorded, and this diagnostic applies no optimizer or precision repair.

Different batch shapes can round contractions and return reductions differently despite representing the same trajectory. For local categorical credit g_i=pi_i(Q_i-mean_pi(Q)), a return error bounded by delta gives score-credit error at most2pi_i delta, before the score Jacobian pullback and its own rounding. A large common loss can amplify cancellation relative to a tiny advantage. A detached baseline leaves exact credit unchanged, but its numerical benefit must be measured before any fitted-rule change.

Theory117;13.556s/374828KiB, four contracts. Full gradient vectors and actual factual/shadow winner histories are preserved in the source-hashed vectors artifact. Total audit arithmetic/traffic/energy are unknown, not zero. No trained text8, DEV/test or Adam step. This small rounding floor does not establish a cause of underfitting or absent deep features; larger utility, horizon, exposure and resource questions remain separate.

# Appendix B. Production causal depth8 replay driver: numerical admission complete

| L8/p4 family/credit | Whole fit GF | Fit MF/target | Infer MF/char | Shadow lanes |
|---|---|---|---|---|
| private/teacher | 0.002514 | 0.052384 | 0.010616 | 0 |
| private/factorized | 0.002687 | 0.055975 | 0.010616 | 0 |
| private/replay | 0.364215 | 7.587809 | 0.010616 | 1536 |
| depth/teacher | 0.002400 | 0.050005 | 0.010616 | 0 |
| depth/factorized | 0.002573 | 0.053596 | 0.010616 | 0 |
| depth/replay | 0.364101 | 7.585430 | 0.010616 | 1536 |

All SIX actual private/shared teacher/factorized/full-replay drivers pass bitwise interrupted/resumed model/Adam/private state/cursor/RNG/counter/work recovery, including an UNTRACED third credit chunk. Each row is the SAME48 text8 fitting targets, one pass, credit16/U16/lr.002/warmup32,33 disjoint DEV characters. Every stage covers actual shadows/backward/normalize/clip/Adam; the accumulator includes backward in its combined work bucket. None of these tiny correctness fits is a language quality comparison.

## Full corrected language replay: avoid the redundant winning shadow

| p4/L8 private replay | Fwd+back GF | Fwd+back MF/target | Shadow lanes |
|---|---|---|---|
| full | 0.004387707 | 1.462569 | 96 |
| reuse | 0.002269611 | 0.756537 | 48 |

Independent AWS winner-reuse proof: four private/shared one/three target double cases agree for EVERY gradient within3.26e-15 absolute error and have bitwise factual logits/state/end RNG. The second table uses the SAMEthree-target denominator and includes only forward+backward:48.2734% less work, optimizer/inference/traffic/RNG/energy excluded and not zero. Its subsequent TWO actual optimized driver recovery contracts pass, with768lanes/12288shadowevents per48-target fit. No candidate sampling or inference substitution.

All columns use2FLOPs/MAC+unit-special and consistent denominators within their table; whole actual fit and per-target work appear together. The isolated three-target audit is explicitly separate from full 48-target driver fits. Original teacher10M checkpoints were preserved at73728targets/288updates each for exact continuation. Original failed object-comparison test and source-scoped coordinator recovery remain historical; no learned weights were discarded. Source-backed AWS findings/manual report history are retained alongside these pages.

# Appendix B. AWS deep language smokes: same learning with half the replay work

| L8/p16 family/credit | Final DEV BPC | Whole fit GF | Fit MF/target | Infer MF/char |
|---|---|---|---|---|
| private/teacher | 4.735412 | 0.463355 | 0.452495 | 0.097376 |
| private/factorized | 4.737501 | 0.476965 | 0.465786 | 0.097376 |
| private/replay | 4.730272 | 69.349421 | 67.724044 | 0.097376 |
| private/reuse | 4.730272 | 34.913923 | 34.095628 | 0.097376 |
| depth/teacher | 4.706595 | 0.461288 | 0.450477 | 0.097376 |
| depth/factorized | 4.698587 | 0.474900 | 0.463769 | 0.097376 |
| depth/replay | 4.695411 | 69.347356 | 67.722027 | 0.097376 |
| depth/reuse | 4.695411 | 34.911858 | 34.093611 | 0.097376 |

ALL eight completed real causal text8 learning/RSS admissions: 1025 observed FIT characters/1024 next-character targets, one pass, 129 disjoint DEV characters at90M/128targets, seed7, p16/L8/H2/pool2, credit16/U256/lr.002/warmup4096/clip1. Same chronological persistent sparse state, native computational delays, key/value separation and receiver maps. Private54907/shared22687parameters; 32available/32scored/16selected receiver writes pertarget. Original teacher, factorized clock/content and full-write choice credit are different learning estimators of this integrated model.

private: original→reuse BPC difference +3.44e-07; counted whole-fit work -49.655%; measured wall 554.1→379.2s. depth: original→reuse BPC difference -3.44e-07; counted whole-fit work -49.657%; measured wall 548.4→374.3s. Full replay pays32768shadow lanes/524288 shadow events; reuse pays16384/262144, retaining ALLalternative write returns and unchanged sparse inference. All fitting stages are counted together; inference is the same warm selected-value native prefix including next-character loss. The measured wall comparison spans separate same-host admitted jobs, not a controlled throughput or energy benchmark.

Every smoke lowers its initial DEV BPC and stays below1GBRSS. Private initial5.209543, shared5.134375; replay/reuse final scores agree within3.44e-7 BPC. These small fits demonstrate learning and implementation admission. They do not satisfy the user's10M character quality protocol, establish generalization advantage against counts/Transformers, or imply full replay's total training work is competitive with the much cheaper teacher/factorized controls.

Prioritized matrix aws_language_winner_matrix_20261003T014100Z: exact original10M teacher continuations, private and depth-shared optimized full-write replay10M, and factorized10M controls as slots permit. Long outcomes remain PENDING; no predicted BPC is entered. Three bounded one-thread CPU slots on the authorized AWS host, host+slot locks/RSS watchdogs/min8GiB and serialized publication. All earlier positive/failed depth8 gesture evidence and original full replay learning rows are retained. CPU arithmetic estimates include clock simulation/gradients/optimizer; DEV passes/RNG/traffic/energy separate, no physical cost projection or supremacy claim. Original teacher recovery may discard at most4095 uncheckpointed targets each (<=.041% extra relative to10M); exact discarded work is unknown. Original1M initial DEV versus new1025-character initial diagnostics also makes startup wall unequal; final1M DEV is matched.

# Appendix B. Cached causal prefixes: equivalent credit with fewer operations

| Family/arm | Whole fit GF | Fit MF/target | Infer MF/target | Shadow lanes | Shadow events |
|---|---|---|---|---|---|
| private/full | 0.069989 | 17.497187 | 0.095498 | 128 | 512 |
| private/winner | 0.036360 | 9.090061 | 0.095498 | 64 | 256 |
| private/cached | 0.023758 | 5.939437 | 0.095498 | 64 | 160 |
| depth/full | 0.069473 | 17.368150 | 0.095498 | 128 | 512 |
| depth/winner | 0.035844 | 8.961023 | 0.095498 | 64 | 256 |
| depth/cached | 0.023242 | 5.810399 | 0.095498 | 64 | 160 |

SAME synthetic FOUR targets, p16/L8/H2/pool2 represented weights, nonempty private state, entering RNG and targetweighted Adam update. All forward/shadow backward, cache copies/stacking, normalization, clipping and Adam stages have complete operator accounting. Both native families keep32 available receivers, 32 scored keys and16 selected updates per target. No quality/data comparison or T4-to-T16 FLOP projection. Inference uses selected values; cold native prefix accounting differs from AWS warm-prefix-plus-loss accounting.

Full enumeration repeats the factual prefix for every forced alternative. Winner reuse already removes the factual winning shadow. The new causal cache also saves detached pre-token memory/arrivals/context and RNG, then evaluates only each losing route's remaining suffix. Same complete future loss returns and every race alternative remain. This changes execution work, not the conditional-credit objective or sixteen-token detachment boundary.

Theory118's36 contracts cover private/shared p4/L8 U1/U2/U4, empty and nonempty memory, every causal snapshot and ALL losing suffixes against full-prefix shadows, EVERY gradient against old batched AND independent sequential enumeration, primal/private-state/RNG identity and exact pending gradient/Adam recovery. Theory119 adds productionT16 double EVERY-gradient equivalence and actual T4 complete-work admission for all three arms.

Contracts 29.071s/390804KiB; resource campaign 82.818s/431104KiB. Snapshot tensors and RNG are charged as bytes; Python metadata, graph memory, traffic/RNG/energy remain separate. Total campaign fitting work is unknown, not zero. Original winner-reuse tight-coordinate failures and precision audit are retained.

# Appendix B. Cached-prefix prototype: CPU wall regression prevents promotion

| Family | Full seconds | Winner seconds | Cached seconds | Cached/winner | Double cache bytes |
|---|---|---|---|---|---|
| private | 1.654061 | 1.221224 | 2.578835 | 2.112 | 154744 |
| depth | 1.631635 | 1.217643 | 2.615722 | 2.148 | 152568 |

T16 FLOAT32 learning-step wall, three fresh serial replicates per arm with rotated order, SAME inputs/noise/model/Adam/warmup. The table reports medians; all individual times are saved. Timing includes accumulation, one equal gradient-vector diagnostic extraction, normalization, clipping and Adam; it excludes initialization/DEV and is not an energy measurement. Cache-byte column is the separate T16 DOUBLE admission, not peak float32 memory.

The grouped cache is slower despite fewer counted T4 operations. AtT16 the original/reuse/cache shadow events are8192/4096/2176, while lanes are512/256/256. Cached suffix grouping expands sixteen batched shadow token iterations to136 smaller iterations. Repeated stacking, Python/operator launch overhead and smaller batches are plausible causes, not yet isolated by a profiler. Fewer event simulations do not guarantee lower wall, traffic or energy.

Every float32 replicate's factual logits/private state/endRNG matches exactly across arms, global gradient error stays within3e-5 and actual Adam displacement error within.005. The production double comparison matches EVERY gradient to both older programs. These newly fixed global budgets do not replace Theory116's failed coordinate gate. No quality benchmark was trained or selected.

Decision: retain exact winner reuse in the separately owned AWS10M matrix; do not promote this grouped cache. A compact executor for variable-length live shadow lanes is a future contract requiring per-lane chronology/noise identity. It must demonstrate actual wall and resource benefit before another long fit.

Theory118/119. All six T4 audited updates, eighteen T16 timed updates and six double gradient fits are paid diagnostic work. Inference and counterfactual coverage remain unchanged. Negative wall findings stay beside positive arithmetic savings; ten-million-character quality remains pending.

# Appendix B. Progressive depth growth: preserve the actual parent computation

| D6 layer | Legacy gain | Preserved gain | Construction origin |
|---|---|---|---|
| 1 | 0.250000000 | 0.353553391 | inherited D2 |
| 2 | 0.250000000 | 0.353553391 | inherited D2 |
| 3 | 0.250000000 | 0.250000000 | inherited D4 |
| 4 | 0.250000000 | 0.250000000 | inherited D4 |
| 5 | 0.204124145 | 0.204124145 | new D6 |
| 6 | 0.204124145 | 0.204124145 | new D6 |

Concrete bug: residual unit.gain is a Python float outside state_dict. Direct D2->D4 initializes the old layers at.353553 and new layers at.25. Legacy D4->D6 resets ALL old layers to.25, reducing the oldest two nonlinear amplitudes29.29% before fitting. Tensor-only checkpoint copying cannot preserve this part of the computation. Existing direct D2->D4 completed results remain valid for their original variant.

The new sibling reconstructs ordinary, legacy-grown and new parents from source-checked lineage, then preserves their ACTUAL gain vector. Existing historical resets are reconstructed as historical resets. Every checkpoint/result binds ancestor bytes and records gains, including interrupted snapshots. Unknown construction, missing/cyclic lineage, changed sources and contradictory state/metadata refuse admission.

Twelve completed contracts: direct D2->D4 bitwise old/new logits, EVERY gradient, persistent state and caller RNG; heterogeneous gains and old tensors retained at D6; artifact reload through D8; actual activated batched factory and interruption annotation; historical reset interpretation; invalid lineage rejection and parent/source immutability. Tiny synthetic numerical cases, no fitted quality.

Closed new sigmoid gates remain at bias-20, a plasticity concern requiring actual gate/branch/update measurements. Channel and clock paths remain live. Fresh Adam, extra passes, positive race delays and shifted parent RNG remain confounds. The fix preserves gains; it does not claim exact parent-function preservation or useful new nonlinear features. A matched shallow continuation is still required.

Theory121;1.560s/261216KiB. Original growth driver/queues/results frozen. Pending external legacy D6 queue is affected; use the new sibling with a unique tag after its owner checks parent artifacts. No temporal/sparse mechanism removed; total contract work/traffic/energy unknown, not zero.

# Appendix B. Actual Adam updates: clipping scale and moment history differ

| Protocol/FIT offset | Raw norm | Cap1 step | Cap4 step | Surrogate cosine1/4 | Anchor delta1 | Anchor delta4 |
|---|---|---|---|---|---|---|
| D2 fine/0 | 5.8978 | 0.08956 | 0.14076 | 0.1652/0.4605 | -0.04537 | -0.13070 |
| D2 fine/16 | 2.7194 | 0.09672 | 0.13851 | 0.2205/0.3975 | -0.00149 | -0.07285 |
| D2 fine/32 | 3.1508 | 0.09686 | 0.14474 | 0.2520/0.4224 | +0.03192 | +0.06702 |
| D2 fine/fresh | 2.6687 | 0.37366 | 0.37373 | 0.3535/0.3535 | -0.19458 | -0.19458 |
| D4 coarse/0 | 5.6617 | 0.12771 | 0.17440 | 0.0905/0.3155 | +0.00487 | -0.07074 |
| D4 coarse/16 | 4.3436 | 0.13583 | 0.19905 | 0.1510/0.3455 | -0.08828 | -0.30569 |
| D4 coarse/32 | 3.6024 | 0.12242 | 0.16666 | 0.0545/0.2973 | +0.02082 | -0.01406 |
| D4 coarse/fresh | 3.3842 | 0.50187 | 0.50283 | 0.3362/0.3355 | -0.24643 | -0.24643 |

Within each row, SAME frozen parameters, normalized actual driver gradient, Adam moments and FIT inputs. Step columns are full parameter displacement L2; anchor delta is NLL on fixed independent FIT96..111, not DEV/test. Three trained windows FIT0..15/16..31/32..47 and one fresh first step per protocol; six discarded cap/LR forks per window. Quarter LR and no-cap outcomes plus every per-block/vector are saved.

Fresh Adam nearly cancels constant gradient rescaling: cap1/4 step norm differs.019% in D2/.190% in D4 despite substantial gradient shrinkage. With trained moments, cap4 changes direction and increases D4 step norm1.36-1.47x, rather than fourfold. Three D4 frozen FIT losses and anchors improve more at cap4. This changes CURRENT gradient weight against cap1-trained moments; it is not a history trained under cap4. The eight-pass cap4 failures remain valid.

Protocols differ: D2 fine20 seed6/eight-pass native teacher versus D4 coarse4 seed7/four-pass factorized pilot. No causal cross-depth comparison. D2 preprocessing hash reproduces; D4 ORIGINAL statistic hash differs from recomputation on this host, explicitly recorded. D4 uses controlled CURRENT FIT inputs at genuine paired online weights/moments. Exact original-input replay remains an artifact gap. Its frozen original batched-kernel bytes are archived and loaded.

Theory120/122;100.552s/382576KiB, five checks. Adam fresh-step formula and constant-history scaling pass. Directional cosine uses the declared SURROGATE, not exact hard branch derivatives; finite loss is observed. Four failed attempts remain archived. Total diagnostic FLOPs/traffic/energy unknown, not zero; no cap schedule or benchmark improvement established.

# Appendix B. Added message branches: float32 silence and Adam epsilon

| New bias | Layer | Mean gate | Visible delta % | Gate step L2 | Output step L2 | Gate active<eps % |
|---|---|---|---|---|---|---|
| -20 | 3 | 2.28e-09 | 0.265 | 1.4e-05 | 1.96e-05 | 100.00 |
| -20 | 4 | 2.22e-09 | 0.251 | 1.5e-05 | 2.74e-05 | 100.00 |
| -8 | 3 | 0.000371 | 99.958 | 0.0777 | 0.0846 | 15.53 |
| -8 | 4 | 0.000361 | 99.963 | 0.075 | 0.0829 | 21.05 |
| -4 | 3 | 0.0198 | 100.000 | 0.0975 | 0.0955 | 0.28 |
| -4 | 4 | 0.0193 | 100.000 | 0.0971 | 0.0953 | 0.37 |
| 0 | 3 | 0.509 | 100.000 | 0.0988 | 0.096 | 0.09 |
| 0 | 4 | 0.502 | 100.000 | 0.0987 | 0.0959 | 0.00 |

Prespecified -20/-8/-4/0 gate-only interventions at identical grown p16/H2/pool2 D4 from a saved trained D2 parent. SAME sixteen FIT gestures0..15, exact fine transform, fixed common race noise,

one fresh actual normalized clip1 Adam.003 update per arm. Visible delta is the fraction of candidate FP32 proposals differing from incoming content. Steps are actual stored-parameter displacement norms.

Legacy -20 gates average roughly2.2e-9. About99.7% of added candidate message contributions round away when added to incoming float32 values. ALL active gate/output gradients are below Adam eps after clipping; mean gate updates receive only about.007% of the unattenuated fresh sign step. Their aggregate displacement is about 1.4-1.5e-5, versus.097 at -4. This directly supports a practical nonlinear-message plasticity problem in this growth construction.

Added input maps remain live: -20 raw gradient norms.046/.139 and Adam displacement about.096 per layer, via persistent memory key/timing paths. Channel mixes and clocks also remain active. Therefore added depth is not entirely frozen, and its old accuracy gain cannot be attributed exclusively to either old layers or new channels without lesion/continuation controls.

At -4 all measured candidate contributions are visible and gate/output steps nearly escape epsilon attenuation. This is a GATE-ONLY counterpart to the other host's pending §410; its full scratch initialization/transport/fit protocol differs. All four one-step FIT losses decrease, including -20. No DEV/test, tuning or benchmark advantage, and no inference-cost gain established.

Theory123/122;6.142s/370288KiB. All four instrumented logits/EVERY gradient bitwise nest plain computation; fresh-step formulas, parent/source/RNG/kernel immutability pass. Four independent updates/64 target exposures plus verification work; total FLOPs/traffic/energy unknown, not zero. Reuse the owner's initialization comparisons before proposing another large fit.

# Appendix B. Conditional race variance: actual parameters and Adam updates

| Depth | k | k/R | Exact combined MSE ratio | Mean gradient cosine | Mean Adam rel error | Mean Adam cosine |
|---|---|---|---|---|---|---|
| 2 | 8 | 0.095 | 0.01793 | 0.99122 | 0.7707 | 0.70207 |
| 2 | 32 | 0.381 | 0.003068 | 0.99833 | 0.6027 | 0.81773 |
| 4 | 8 | 0.048 | 0.08463 | 0.96108 | 0.9359 | 0.56121 |
| 4 | 32 | 0.190 | 0.01798 | 0.99159 | 0.7896 | 0.68761 |
| 6 | 8 | 0.032 | 0.3051 | 0.90586 | 0.9288 | 0.56706 |
| 6 | 32 | 0.127 | 0.06876 | 0.96771 | 0.8262 | 0.65810 |

Initialization-only double-precision audit: same represented float32-initialized p16/H2/pool2 weights promoted to double, seed7, FIT examples0/1, fixed common history noise. Factories change gain, parameter count and RNG consumption with depth; this is not an isolated causal depth perturbation. FIT-only normalization loads examples0..15 but only0/1 enter this probe; no DEV/test arrays read.

For every legal race, cache its normalized full factual-score parameter VJP of sum pi times detached alternative suffix losses. Its sum matches both the all-race objective and original driver. A fixed sampled8 cached sum matches a separately differentiated subset objective using the same returns; it does not independently execute the sampled8 driver. A tiny actual native R4/k2 case enumerates all six subsets to verify unbiasedness and covariance.

Exact trace covariance for uniform k without replacement is R(R-k)/(k(R-1)) times the sum of squared centered per-race parameter vectors; independent episode subset covariances add. The displayed MSE divides by squared FULL combined factual-plus-route gradient, not only a small canceling route mean. Sixty-four cached draws per row give descriptive cosine/update means, not fitted quality or Monte Carlo estimates of the exact covariance.

Fresh Adam transforms include actual clip1 normalization and epsilon1e-8; full and sampled updates are independently checked against two actual discarded Adam forks per depth. Results do not describe trained moments, convergence or heldout improvement. Kernel, original parameters and caller RNG are preserved. Sparse native inference is unchanged.

Five contract groups;292.571s/866372KiB. Accounting correction beside original artifact scope: every alternative forward bank is evaluated TWICE, once for vectors and once for full-driver equivalence; shadow_lanes/events in each case counts only the first bank. Tiny contracts add their own work. Cached draws add vector/Adam computation; total diagnostic FLOPs/traffic/energy unknown, not zero. Full per-race vectors are a228MiB generated local artifact with an immutable SHA and reproducible source/queue; report tables use the completed JSON. Local bank absent on this rendering host; vector-byte verification not rerun.

# Appendix B. Sampled credit: finite FIT predictions after actual Adam forks

| Depth/credit | Forks | Same FIT NLL delta | Anchor NLL delta | Fresh-noise anchor delta | Anchor KL vs full | Fresh-noise KL vs full |
|---|---|---|---|---|---|---|
| D2 full | 1 | -0.80321 | +0.07646 | +0.06996 | 0.00000 | 0.00000 |
| D2 factorized | 1 | -0.80844 | +0.07066 | +0.07961 | 0.00051 | 0.00097 |
| D2 k8 | 8 | -0.78296 | +0.06691 | +0.07583 | 0.00076 | 0.00116 |
| D2 k32 | 8 | -0.78837 | +0.07762 | +0.07689 | 0.00044 | 0.00070 |
| D4 full | 1 | -1.26088 | +0.14452 | +0.11285 | 0.00000 | 0.00000 |
| D4 factorized | 1 | -1.27601 | +0.13249 | +0.10531 | 0.00092 | 0.00089 |
| D4 k8 | 8 | -1.20494 | +0.09725 | +0.09090 | 0.00899 | 0.00981 |
| D4 k32 | 8 | -1.22689 | +0.11830 | +0.10728 | 0.00241 | 0.00212 |
| D6 full | 1 | -1.41862 | +0.46718 | +0.46173 | 0.00000 | 0.00000 |
| D6 factorized | 1 | -1.45646 | +0.50830 | +0.55700 | 0.00460 | 0.00640 |
| D6 k8 | 8 | -1.36269 | +0.43364 | +0.46939 | 0.02407 | 0.02747 |
| D6 k32 | 8 | -1.41032 | +0.45229 | +0.49768 | 0.01226 | 0.01884 |

Same frozen initialization/parameters/gains and normalized two-example gradients as Theory124. Per depth: factual-only and full-choice controls, plus the FIRST EIGHT archived draws for k8 and k32; no selection by outcome. Every fork executes fresh clip1 Adam.003 and verifies its actual stored displacement. Parameter ordering and initial original-noise FIT NLL reproduce the source exactly.

NLL deltas are relative to each depth's unchanged model under the SAME evaluation noise; negative means improvement. Same FIT uses0/1, anchors are disjoint unused FIT2..15 with the original FIT-only normalization. Fresh-noise columns use a fixed second whole-history draw171324; other columns use original171323. Anchor examples are not an IID or heldout split. Prediction KL is from full-credit fork to each arm under matching noise.

All54 forks improve their two fitting examples under both noise draws, while ALL worsen these disjoint FIT anchors. AtD6 full-credit anchor NLL increases.46718/.46173 versus.14452/.11285 atD4. The tiny gradient batch covers classes0/1, whereas the anchors contain other classes and adjacent subject recordings. This exposes a first-step fitting-versus-anchor tradeoff, not a representative minibatch or trained generalization diagnosis.

Eight-fork rows show means, not a selected best fit. All54 actual outcomes, losses, logits and subset indices remain in the artifacts. Large parameter-step differences do not automatically mean worse predictions: the finite forward is the relevant functional check. Conversely, a favorable first step cannot demonstrate convergence, quality advantage or rescue of a trained deep model.

The conditional-choice/factorized-time surrogate need not descend every literal fixed-noise realized loss; two noises are a bounded diagnostic, not complete expected-risk integration. Detached losing contents, persistent addressed state, key/value separation and computational races remain unchanged. No DEV/test array, dense fit, architecture substitution or trained moment history was used.

Theory125;four contract groups,20.616s/576664KiB. 54 executed optimizer forks,1824 prediction-target evaluations; paid parent gradients are reused with no new counterfactual bank. Total diagnostic FLOPs/traffic/energy unknown, not zero. Other-host live-branch initialization comparisons and the AWS streaming10M quality matrix remain the integrated priorities.

# Appendix B. Actual training batch: contracted parameter projections

| Depth | Batch | k | Combined MSE ratio estimate | Empirical SE | Relative SE % | Sign probes |
|---|---|---|---|---|---|---|
| 4 | 16 | 8 | 0.080239 | 0.003471 | 4.33 | 32 |
| 4 | 16 | 32 | 0.017051 | 0.000738 | 4.33 | 32 |
| 6 | 16 | 8 | 0.147045 | 0.006484 | 4.41 | 32 |
| 6 | 16 | 32 | 0.033145 | 0.001462 | 4.41 | 32 |

p16/H2/pool2, initialized seed7, SAME original fine FIT0..15 transform and fixed common race noise. Each full16-example episode graph retains computational delays, temporal races, private addressed state and separate keys/values. No DEV/test arrays, optimizer step, fit quality or inference saving. Different depth factories are a configuration ladder, not an isolated causal depth perturbation.

Pool2 conditional choice credit equals pi0*pi1*(Q0-Q1)/B times the complete parameter Jacobian of score0-score1. An artificial score cotangent gives that Jacobian times a random parameter-sign vector through reverse-over-reverse pullback. This differentiates cotangents, not a physical stochastic Hessian. Detached alternative returns and the native factorized clock/value backward remain unchanged.

THREE directions per D2/D4/D6 reproduce EVERY projected race contribution against Theory124's exact vectors, maximum error1.38e-14. Main estimates use32 independent Rademacher directions with no1/sqrt(parameter-count) scaling. Exact finite-population sampling covariance is projected and centered within each legal episode; its normalized trace estimate is unbiased. All probes are saved; empirical SE is descriptive, with no guaranteed95% interval. Worst-case relative RMS bound25%.

At actual B16, k8 relative raw-noise RMS is about.283/.383 for D4/D6. Compared with B2 exact MSE.08463/.30506, D4 scarcely changes while D6 about halves. Sampling interference remains measurable in this initialization history. Larger k reduces raw variance, but does not establish Adam-update precision, trained causality, heldout gain or a cure; the finite functional forks retain mixed anchor behavior.

Decision: reuse owned live-gate and gain-lineage comparisons and the original streaming AWS10M quality matrix. No sampling sweep admitted. The90M configuration-selection correction uses DEV metrics, preserving the old written test-based rule beside its correction; completed test scores remain reporting-only.

Theory127/128;56.096s/629624KiB. 13440 full shadow lanes/282240 shadow events,64 main mixed pullbacks plus9 contract projections and factual/full backwards; total diagnostic FLOPs/traffic/energy unknown, not zero. Randomized trace estimation is a known primitive (Avron/Toledo2011); our native conditional-covariance application is contracted, not a novelty or equivalent-work wall-speedup claim.

# Appendix B. Integrated depth4 content learning: paid FIT-only admission

| Credit | Initial FIT NLL | Final FIT NLL | Initial anchor | Final anchor | Learn gate |
|---|---|---|---|---|---|
| choice only | 2.674371 | 2.328692 | 2.604848 | 2.469267 | True |
| joint content choice | 2.674371 | 2.340674 | 2.604848 | 2.470372 | True |

| Model/credit | Data / passes | Eval split / NLL | Whole fit GF | Fit MF/target | Infer MF/target |
|---|---|---|---|---|---|
| choice only | 24 / 2 | FIT-anchor / 2.46927 | 0.039905 | 0.831357 | 0.162128 |
| joint content choice | 24 / 2 | FIT-anchor / 2.47037 | 0.068115 | 1.419073 | 0.162128 |
| Saved p16 D4 factorized | 984 / 8 | DEV / 1.13027 | 34.347614 | 4.363264 | 1.068283 |
| Saved p16 D4 full replay | 984 / 8 | DEV / 1.02063 | 3815.206646 | 484.655316 | 1.068437 |

Both pass the fixed learning gate, but joint content is worse by 0.011983 FIT NLL and 0.001105 anchor NLL, with 1.707x fitting work. No promotion or retuning. Saved factorized/full replay fitting work per presentation is 3.07x/341.53x the joint smoke; unequal width/data/quality prevent a superiority claim.

SAME fresh p4/D4/H2/pool2, original fine21-event FIT0..23, fixed disjoint FIT24..31 anchors, two passes/48 presentations/12 Adam updates per arm, B4/lr.003/clip1. Identical initialization/order/sites/common noise, ordinary gates, no growth or pretrained weights. Anchors are adjacent FIT recordings, not IID or official DEV/test. Every update is fully traced; source contracts include actual real-depth4 EVERY-gradient decomposition and BLk1 identity.

Each smoke keeps16 available private receivers, eight selected updates and16 key scores per event;21 events means168 selected updates/336 scores per target. Full-prefix shadows96 lanes/2016 events per fit are charged, including ALL live-branch backwards in the joint arm. Objective normalization is charged inside forward; no second scaling. Native inference is identical in structure and counted on the same32-target boundary.

The saved full-data rows retain stronger quality, other widths and7872 presentations, with first/last-window work estimates. Smoke work is exact over48 presentations. EVERY table column uses common units and presentation denominators; unequal quality/data make raw work gaps admission evidence, not comparable-quality superiority. Evaluation/verification and failed-attempt work are separate unknown work, not zero.

Original042000Z run hit its180s guard after choice-only completed; joint was incomplete. Preserve original source/protocol/logs and48-96 failed fitting-presentation bound. Retry changes only the timeout to420s based on measured tracing cost; same data/passes/settings, no loss-based tuning or extension. Incomplete scores are excluded from this paired table.

Theory130;230.670s/469892KiB. Tiny single-seed learning/resource admission only: no useful deep-feature attribution, batch variance reduction, heldout confirmation or supremacy. Existing integrated live-gate comparisons and AWS streaming10M remain priority; no active source replaced.

# Appendix B. Trained batch credit: empirical variance and real Adam moments

Recover the previous choice-only native p4/D4/H2/pool2 fit with its EXACT twelve steps,48 presentations, FIT0..23, normalization, sites/noise and clip1 Adam.003. Every saved per-step factual/conditional loss, final FIT/anchor prediction and parameter-group movement agrees. Reconstructed weights AND actual Adam history are archived and hashed. This is reconstruction, not another benchmark or seed.

| Quantity | Factual content trace | Conditional content trace | Joint/factual | Leave-one-pair-out |
|---|---|---|---|---|
| raw | 7.86872441 | 7.85248081 | 0.997936 | 0.9967..1.0003 |
| clipped | 0.395247263 | 0.396003281 | 1.001913 | 1.0002..1.0029 |
| update | 0.000346709069 | 0.000345469077 | 0.996424 | 0.9958..0.9974 |

| Credit | Content trace | Choice trace | 2x cross trace | Total raw trace |
|---|---|---|---|---|
| factual | 0.320772 | 7.571728 | -0.023776 | 7.868724 |
| joint | 0.300850 | 7.571728 | -0.020097 | 7.852481 |

Here content variance falls 6.21%, but choice credit alone is 96.23% of total raw trace. Total raw variance falls only 0.206%, while clipped variance rises and real-Adam variance falls only 0.358%. Exposure helps a small term in this restricted sampled-one-site teacher.

SAME trained state and original FIT0..3, B4,16 paired prespecified fresh-noise/uniform-site draws. Extract factual content C, conditional live content Cbar and the SAME R-scaled choice A. Shared race noise across episodes is retained. Raw vectors are actual float32 C+A versus Cbar+A. Full parameter covariance is measured; content/choice cross terms are signed. Zero gradients and disconnected None gradients remain distinct.

Each pair forks the SAME recovered warm moments through actual clip1 Adam;32 updates are discarded. Raw/clipped/update vectors and parameter-group statistics are retained. This tests the nonlinear optimizer, rather than equating its update with a raw gradient or fresh sign step. Sample traces and leave-one-pair-out ranges are descriptive;16 draws do not provide guaranteed intervals or exact expected-risk gradients.

Contracts compare EVERY parameter in double with isolated native objectives, then float32 gradients and real updates; serialized next-update recovery, causal fixed inference, source/data/state/moment/RNG preservation and operator coverage pass. No episode-noise independence assumption is inserted into the actual batch. Earlier per-episode conditional-mean theorem and its batch-covariance limitation remain explicit.

Theory132; 65.157s/430804KiB. One tiny restricted trained state, not all depths or the AWS language failure. No heldout nomination, convergence guarantee, larger fit or superiority claim.

# Appendix B. Trained content forks: paid work and fixed FIT predictions

| Credit | Isolated step GF | Fit MF/presentation | Infer MF/target |
|---|---|---|---|
| factual | 0.003325429 | 0.831357 | 0.162245 |
| joint | 0.005835553 | 1.458888 | 0.162245 |

| Variance quantity | Joint/factual trace | Step work ratio | Variance x work ratio |
|---|---|---|---|
| raw | 0.997936 | 1.754827 | 1.751205 |
| clipped | 1.001913 | 1.754827 | 1.758184 |
| update | 0.996424 | 1.754827 | 1.748551 |

| Credit | Split | Initial NLL | Mean delta NLL | Improved draws | Delta range |
|---|---|---|---|---|---|
| factual | fit | 2.030324 | -0.035250 | 16/16 | -0.04004..-0.02475 |
| factual | anchor | 2.469267 | -0.008652 | 14/16 | -0.01999..+0.00297 |
| joint | fit | 2.030324 | -0.035248 | 16/16 | -0.04004..-0.02474 |
| joint | anchor | 2.469267 | -0.008648 | 14/16 | -0.02006..+0.00303 |

Only TWO isolated actual fitting windows, four presentations each, are charged in the work table: normalized forward/loss, all forced branches, live backward where applicable, clipping and restored historical Adam. Native inference uses the SAME twelve targets. One special-function evaluation counts as one. This is not the whole ensemble campaign or a new whole fit. The prior page retains complete48-presentation fits and saved full-data references with consistent units/denominators.

Same16 available private receivers,8 selected state updates and16 key scores per event;21 events gives168 selected/336 scored per target. Inference structure and capacity are identical. The live losing-message derivative changes learning work, not available state or candidate discovery support. No inference saving follows from this estimator.

All32 discarded updates are evaluated on fixed FIT0..3 and FIT24..31 anchors under common inference noise314159;384 prediction-target evaluations plus baseline12. Adjacent FIT anchors are

not IID, DEV or test. Every outcome is saved; none selects a configuration. Mean loss changes from one warm update are local effects, not generalization or sustained learning.

Reconstruction48 presentations/12 updates; ensemble64 factual targets,128 full shadow lanes/2688 events and48 component reverse calls; extra two isolated paid updates and two recovery forks, plus admissions/evaluation. Total campaign FLOPs/traffic/energy remain unknown, not zero. Variance-times-work is a declared optimization heuristic, not a convergence theorem for clipping, Adam or nonlinear sparse temporal models.

Preserve the tiny paired-fit negative result and these trained-state measurements together. No long content fit is admitted. Priority remains completed AWS corrected full replay10M and matched teachers/controls; other-host live-gate and calibrated reset-segment language runs use distinct protocols.

# Appendix B. Separate site sampling from native race history

| Sites per episode | Mean site trace | History mean trace | Total raw trace | Site share % |
|---|---|---|---|---|
| 1 | 7.046586 | 0.341525 | 7.388111 | 95.38 |
| 8 | 0.843903 | 0.341525 | 1.185427 | 71.19 |
| 32 | 0.179329 | 0.341525 | 0.520854 | 34.43 |
| 168 | 0.000000 | 0.341525 | 0.341525 | 0.00 |

SAME source-bound12-step trained p4/D4/H2/pool2 and original FIT0..3/B4, FIRST FOUR prior noise histories, with represented weights promoted to DOUBLE. Each history pays complete detached returns for ALL168 races per episode; the original native full-site teacher is the conditional mean. Uniform without-replacement site sampling is the only random axis in each conditional covariance.

At fixed E, v_jr=pi0*pi1*(Q0-Q1)/B times the COMPLETE parameter Jacobian of score0-score1. Site covariance trace S_k=sum_j R_j*(R_j-k)/(k*(R_j-1))*sum_r ||v_jr-mean_r(v_jr)||^2. Full-site sampling has zero such variance. Across histories, mu(E)=factual+all-site credit; Var(G)=E[Var_sites(G|E)]+Var(mu(E)). Native computational clock/content coupling remains in the Jacobian.

Here about95% of estimated k1 total variance is site sampling. k8 reduces estimated total raw trace by about84%. These are FOUR-history descriptive estimates, not an attribution of all deep training failure.32 Rademacher parameter signs per history estimate conditional traces without dimension scaling; all probes saved, empirical SE descriptive, worst-case relative RMS25%, no guaranteed interval or actual Adam variance.

Same TRAINED model short2/1-event contract matches EVERY route VJP under three projection signs, full original teacher and explicit subset mean/covariance. Maximum error4.58e-16. All128 main projected sums match actual full gradients. Float32 factual CE reproduces prior draws exactly; all2688 factual winners agree with double, max logit error2.85e-7. This does not prove forced-branch cross-precision equivalence.

Theory134; 59.326s/390920KiB. Main5376 full-return lanes/112896 events and128 projections; tiny equivalence evaluates returns TWICE,96 lanes/160 events. No optimizer step; total FLOPs/traffic/energy unknown, not zero. Stored gradients/returns/signs/projections retain scope; no policy or benchmark promotion from raw variance alone.

# Appendix B. Existing route coverage: actual fitting cost versus variance

| Sites | Actual step GF | Fit MF/presentation | Infer MF/target | Shadow lanes |
|---|---|---|---|---|
| No choice | 0.002028565 | 0.507141 | 0.162245 | 0 |
| 1 | 0.003327802 | 0.831951 | 0.162245 | 8 |
| 8 | 0.012406626 | 3.101656 | 0.162245 | 64 |
| 32 | 0.043533818 | 10.883454 | 0.162245 | 256 |
| 168 | 0.219918762 | 54.979690 | 0.162245 | 1344 |

| Sites | Raw variance ratio | Actual work ratio | Variance x work |
|---|---|---|---|
| 1 | 1.000000 | 1.000000 | 1.000000 |
| 8 | 0.160451 | 3.728174 | 0.598188 |
| 32 | 0.070499 | 13.081853 | 0.922257 |
| 168 | 0.046226 | 66.085291 | 3.054878 |

FIVE EXISTING BL windows, four targets/one update each, SAME trained weights/history, FIRST noise seed and original sampler. Sampled1/8/32, all168 and factorized no-choice. All forward/loss/return/backward/normalization/clip1/Adam.003 work traced. EVERY traced/untraced next weight/moment and serialized full recovery agrees. All updates discarded.

All five use2379 parameters,16 available private receivers, 168 selected updates/336 key scores per target. Losing full returns still compute candidate keys and values; optimizer work is charged. Native inference uses the same12-target boundary. Whole-step and per-presentation work share units/denominators across ALL rows. No choice changes the mean teacher and is not a k0 point on the same route-variance curve.

With equal R, V(k)=a+b/k; approximately affine C(k)=C0+c*k. The interior variance-work optimum is sqrt(C0*b/(c*a)) when a>0, bounded by1..R. Here it is about6.09 sites; k8 is the best MEASURED heuristic point. This is fixed-state estimation math, not a validated adaptive policy or sustained learning guarantee.

Variance-times-work pairs four-history DOUBLE projected raw variance with one-window FLOAT32 actual work. It is an optimization HEURISTIC, not an actual Adam-variance, convergence or quality guarantee. Broader site support buys less sampling noise but costs additional full returns; the remaining race history floor prevents unlimited benefit. No larger fit is selected from local FIT predictions.

Eleven discarded updates: five traced, five exact untraced verifications and one serialized full recovery. Fixed FIT0..3 and adjacent unused FIT24..31 predictions saved for every arm; not IID, DEV or test. Inference traces/reconstruction/admission/verification/evaluation remain additional work. Total campaign FLOPs/traffic/energy unknown, not zero; measured step work is not the campaign total. Earlier negative content evidence retained.

Theory136; 60.076s/362876KiB. Numerically admitted cost/variance tradeoff only. Temporal races, private sparse state/key-values and counterfactual learning retained. AWS corrected replay10M plus exact teachers/controls and other-host live-gate/calibrated language quality remain priority.